



Common Problems

Ask me anything about this topic

Design Strava

Scaling Writes

Real-time Updates

By Evan King · Published Jul 13, 2024 · medium · Asked at:

Understanding the Problem

🏃🚴 What is Strava?

Strava is a fitness tracking application that allows users to record and share their physical activities, primarily focusing on running and cycling, with their network. It provides detailed analytics on performance, routes, and allows social interactions among users.

While Strava supports a wide variety of activities, we'll focus on running and cycling for this question.

Functional Requirements

Core Requirements

1. Users should be able to start, pause, stop, and save their runs and rides.
2. While running or cycling, users should be able to view activity data, including route, distance, and time.
3. Users should be able to view details about their own completed activities as well as the activities of their friends.

Below the Line (Out of Scope)

- Adding or deleting friends (friend management).
- Authentication and authorization.
- Commenting or liking runs.

Non-Functional Requirements

Core Requirements

1. The system should be highly available (availability >> consistency).
2. The app should function in remote areas without network connectivity.
3. The app should provide the athlete with accurate and up-to-date local statistics during the run/ride.
4. The system should scale to support 10 million concurrent activities.



Below the Line (Out of Scope)

- Compliance with data privacy regulations like GDPR.
- Advanced security measures

Here is how it might look on the whiteboard:

Functional Requirements

1. Start, pause, stop and save activities
2. Track and display stats during an activity
3. View details of past activities for self and friends

Non-Functional Requirements

1. Availability >> consistency
2. Operate in low/no network connection
3. Up-to-date local activity stats (low latency)
4. Scale to 10 million concurrent activities

Strava Non-Functional Requirements

The Set Up

Defining the Core Entities

Let's begin by outlining the main entities of our system. At this point, we don't need to delve into every specific detail or column. We'll address the finer points, such as individual fields, once we have a better overall picture. For now, identifying these key entities will help shape our thinking and provide a solid foundation as we move towards defining the API.

To satisfy our key functional requirements, we'll identify the following core entities:

1. **User:** Represents a person using the app. Contains profile information and settings.
2. **Activity:** Represents an individual running or cycling activity. Includes activity type (run/ride), start time, end time, route data (GPS coordinates), distance, and duration.
3. **Route:** A collection of GPS coordinates recorded during the activity (this could also just be a field on the Activity entity).
4. **Friend:** Represents a connection between users for sharing activities (note: friend management is out of scope, but the concept is necessary for sharing).

In the actual interview, this can be as simple as a short list like this. Just make sure you talk through the entities with your interviewer to ensure you are on the same page.

Core Entities

- User
- Activity
- Route (could be on Activity instead)
- Friend

Strava Entities

The API

We'll define the APIs that allow the mobile app to interact with the backend services.

Our goal is to simply go one-by-one through the core requirements and define the APIs that are necessary to satisfy them. Usually, these map 1:1 to the functional requirements, but there are times when multiple endpoints are needed to satisfy an individual functional requirement or where a single endpoint meets the needs of multiple functional requirements (as is the case with our activity state update endpoint below).

First, we'll define an endpoint that can be used to create a new activity.

```
// Create a new activity
POST /activities -> Activity
```

```
{
  type: "RUN" | "RIDE"
}
```

Next, we'll add an endpoint that can be used to update the state of that activity.

```
// Update the state of an activity
PATCH /activities/:activityId -> Activity
{
  state: "STARTED" | "PAUSED"
}
```



We use PATCH here because we only want to update a subset of the activity's fields. Whereas PUT would require sending the entire resource, with PATCH we only send the fields that are being updated -- a, albeit minor, improvement in efficiency.

In your interview, PUT and PATCH are functionally equivalent. However, you will come across some interviewers who are RESTful API hawks and will argue this point with you, so it's best to be prepared.

When it comes to tracking activity data, we'll add a POST endpoint that is used to update the activity's route and distance.

```
// Add a new location to the activity's route
POST /activities/:activityId/routes -> Activity
{
  location: GPSCoordinate,
}
```

When it comes to sharing activities with friends, we'll say that this happens automatically when an activity is saved, so instead of a endpoint for sharing, we can just update our existing endpoint that updates the activity state to include a "COMPLETE" state.

```
// Mark an activity as complete using the same endpoint we use to update the activity state
PATCH /activities/:activityId -> Activity
{
  state: "COMPLETE"
}
```

To view activities, we'll need a list view that shows either our activities or the activities of our friends. This list will just show basic information about the activity, such as the distance, duration, and date.

```
// Get a list of activities for a user
GET /activities?mode={USER|FRIENDS}&page={page}&pageSize={pageSize} -> Partial<Activity>[]
```



The `Partial<Activity>` type simply means that we are returning a subset of the activity fields. This is done to limit the amount of data we send over the network. This notation is what I use and is derived from TypeScript, but you can use whatever notation you'd like. This is just a personal preference.

When we click on an activity, we'll want to see more details about like the maps with the route and any additional metrics or details (that are out of scope for this question).

```
// Get an activity by id
GET /activities/:activityId -> Activity
```



High-Level Design

Let's address each functional requirement and design components to satisfy them. The best strategy here, especially if it's a question you've never seen before, is to start with the simplest possible system that satisfies the functional requirements and then layer on complexity to satisfy the non-functional requirements during your deep dives.

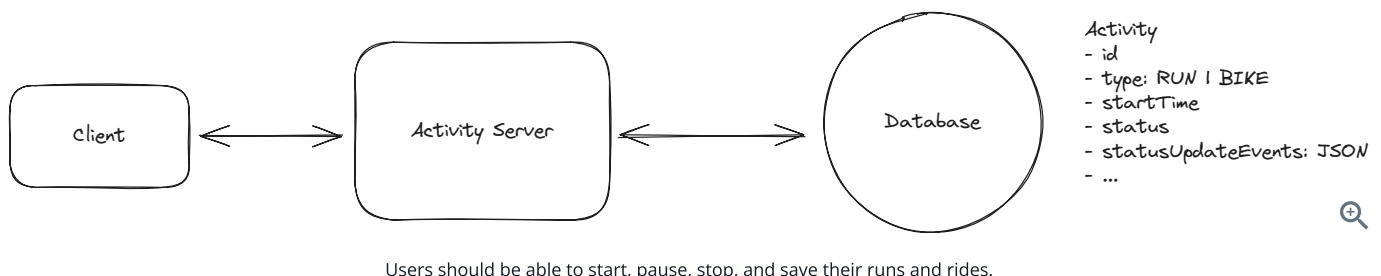


I often get asked, "If I already know the more complex design, should I still start simple?" The answer is nuanced. If you have a well-thought-out, more sophisticated solution in mind, it's perfectly acceptable to present it during the high-level design. That said, it's crucial to strike a balance. Going deep too early can result in you running out of time before you've met all the functional requirements and, thus, not answered the question.

1) Users should be able to start, pause, stop, and save their runs and rides.

We need to design a system that correctly handles both our create activity endpoint and our update activity state endpoint.

We can start by sketching out a very basic client-server architecture:



1. **Client App:** Users interact with the system through a mobile app.
2. **Activity Service:** Handles incoming requests for creating and updating activities.
3. **Database:** Stores information about activities, including route, distance, and time.

A user is going to start by opening the app and clicking the "Start Activity" button:

1. The client app will make a POST request to `/activities` to create a new activity, specifying whether it's a run or a ride.
2. The Activity Service will create the activity in the database and return the activity object to the client.
3. If the user opts to pause or resume their activity, they'll make a PATCH request to `/activities/:activityId` with the updated state and the Activity Service will update the activity in the database accordingly.
4. When the activity is over, the user will click the "Save Activity" button. This will trigger a PATCH request to `/activities/:activityId` with the state set to "COMPLETE".

One interesting part of the design is how to handle the activity time when a user pauses their activity. A naive approach to handling the amount of time elapsed in an activity would be to manage time based on a `startTimeStamp` and then, when an activity is ended, calculate the elapsed time by subtracting the `startTimeStamp` from the current time.

However, this approach would be problematic if a user pauses their activity. The `startTimeStamp` would remain unchanged and the elapsed time would continue to increment, which would be inaccurate.

One common way we could handle this, we can maintain time via a log of status update and timestamp pairs. For example, we could have the following log:

```
[
  { status: "STARTED", timestamp: "2021-01-01T00:00:00Z" },
  { status: "PAUSED", timestamp: "2021-01-01T00:10:00Z" },
  { status: "RESUMED", timestamp: "2021-01-01T00:15:00Z" },
  { status: "STOPPED", timestamp: "2021-01-01T00:20:00Z" }
]
```

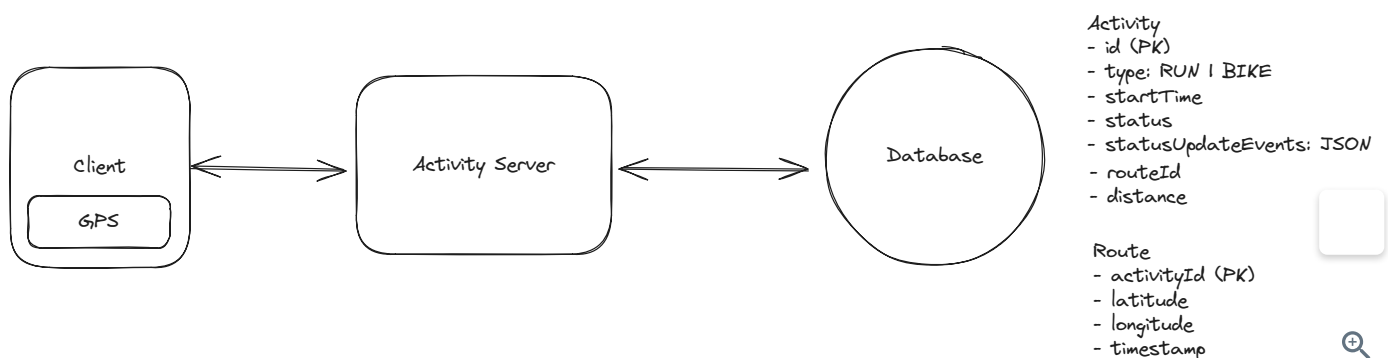
When the user clicks "Stop Activity", we can calculate the elapsed time by summing the durations between each pair of timestamps, excluding pauses. In the example above, the elapsed time would be 15 minutes (10 minutes + 5 minutes).

This may seem like overkill for our simple requirement, but it would allow for natural expansions into a feature that shows athletes when they were paused and for how long, as well as a breakdown of "total time" vs "active time."

2) While running or cycling, users should be able to view activity data, including route, distance, and time.

When the activity is in progress we need to both update the state and display the activity data to the user.

The first, and most important, part of this is to accurately track the distance and route of the activity. We can do this by recording the user's GPS coordinates at a constant interval. We make the assumption that they are using a modern smartphone with GPS capabilities, so we can use the GPS coordinates to calculate the distance using the Haversine formula.



While running or cycling, users should be able to view activity data, including route, distance, and time.

Here is how this would work:

1. The client app will record the user's GPS coordinates at a constant interval, let's say 2 seconds for a bike ride and 5 seconds for a run. To do this, we'll utilize the built-in location services provided by both iOS and

Android:

- For iOS: We'll use the Core Location framework, specifically the `CLLocationManager` class. We can set up location updates using `startUpdatingLocation()` method and implement the `locationManager(_:didUpdateLocations:)` delegate method to receive location updates.
- For Android: We'll use the Google Location Services API, part of Google Play Services. We can use the `FusedLocationProviderClient` class and call `requestLocationUpdates()` method to receive periodic location updates.

2. The client app will then send these new coordinates to our `/activities/:activityId` endpoint.

3. The Activity Service will update the Route table in the database for this activity with the new coordinates and the time the coordinates were recorded.

4. We will also update the distance field by calculating the distance between the new coordinate and the previous one using the Haversine formula.



You absolutely do not need to know the libraries and frameworks to use here to get GPS coordinates. The point of this question is to test your knowledge of system design, not mobile development. However, it is important to understand that most modern smartphones have built-in GPS capabilities and that most mobile apps use the relevant libraries and frameworks to access this data. I've just included them here to give you an idea of how it works in practice.

Astute readers are likely already yelling at their screen as they've realized that we don't need to send these updates to the server, instead handling the logic locally on the client. You're right. We'll get into this in the deep dive. The reality is most of the time I ask this question, candidates overlook this optimization and we get to it later in the interview.

3) Users should be able to view details about their own completed activities as well as the activities of their friends.

Once activities are completed, we should have them already saved in the database and all we needed to do was update their state to "COMPLETE" using the same endpoint we used to update the activity state.

When it comes to viewing activities, we only need to query the DB for the activities with this "COMPLETE" state, while also filtering on the `mode` query param to either show the user's activities or their friends' activities which can be done with a simple WHERE clause.

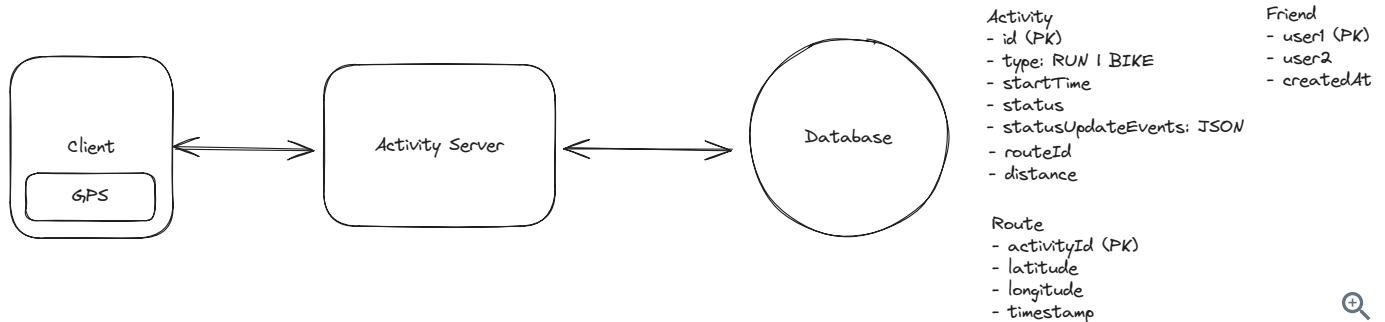
Example queries:

```
-- For mode = USER
SELECT * FROM activities
WHERE state = "COMPLETE"
AND userId = :userId
LIMIT :pageSize OFFSET (:page - 1) * :pageSize

-- For mode = FRIENDS
SELECT * FROM activities
WHERE state = "COMPLETE"
AND userId IN (SELECT friendId FROM friends WHERE userId = :userId)
LIMIT :pageSize OFFSET (:page - 1) * :pageSize
```

Our friends table will have two columns, `userId` and `friendId`. We'll consider relationships to be bi-directional, so for each new friendship we will add two entries to the table, one with the `userId` and `friendId` and the other with the `userId` and `friendId` reversed. By using a composite primary key on (`userId` , `friendId`) and adding an index on `userId`, we can ensure querying for a user's friends is efficient while preventing duplicate friendships.

This initial GET request returns a list of activities with just the basic information and the `activityId` for each activity. When the user clicks on an activity, we can make a second, more detailed GET request using the `activityId` to get the full activity details.



Users should be able to view details about their own completed activities as well as the activities of their friends.

The full flow is thus,

1. User navigates to the activities list page on the app.
2. Client makes a GET request to `/activities?mode={USER|FRIENDS}&page={page}&pageSize={pageSize}` to get the list of activities.
3. The list of activities is rendered in the UI and the user clicks on an activity to view details.
4. Client makes a GET request to `/activities/:activityId` to get the full activity details to render on a new details page.

For showing the map, we can use a combination of the route data and the Google Maps API. We can pass the array of coordinates to the Google Maps API and it will draw a line connecting all the points.

Potential Deep Dives

At this point, we have a basic, functioning system that satisfies the functional requirements. However, there are a number of areas we could dive deeper into to improve efficiency, offline functionality, scalability, and realtime-sharing.

This list of deep dives is, of course, not exhaustive. It is instead an accumulation of some of the most popular directions I've seen this interview take.

1) How can we support tracking activities while offline?

Many athletes will opt to workout in remote areas with no network connectivity, but they still want to be able to track their activities. This brings us to an impactful realization that re-shapes the entire system.

The key insight is that, so long as we don't support realtime-sharing of activities, we can record activity data locally, directly on the clients device and only sync activity data back to the server when the activity completes and/or the user is back online.

Importantly, this actually solves several problems for us:

1. We can track activities without worrying about network reliability on the client side.
2. Even when the client has a fine network connection, we save network bandwidth by avoiding pinging location updates to the server every few seconds (instead, only sending on run completion or pause).
3. Showing accurate and up-to-date activity data to the user is easy as it all happens synchronously on their device.

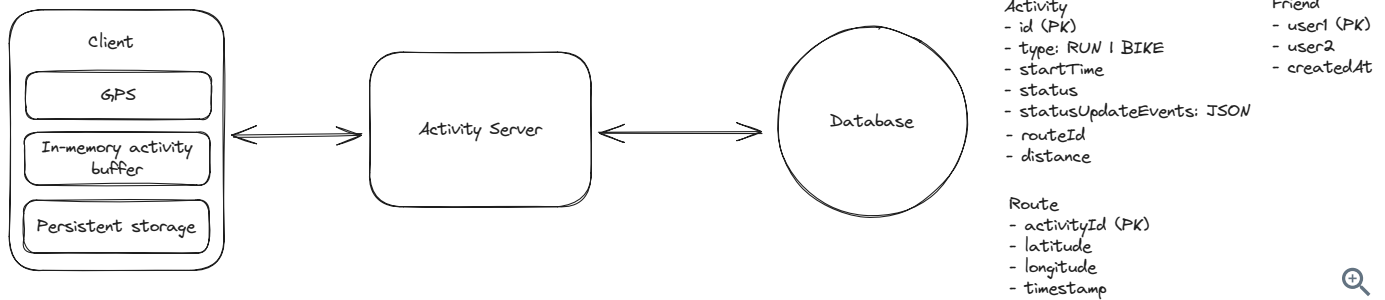


This question highlights an important, often overlooked aspect of system design: the client as an active participant. Modern smartphones and browsers have access to a wide variety of sensors, substantial memory and processing power, and significant local storage. Many candidates overlook the client's capabilities, focusing solely on server-side solutions. However, for systems like Strava, Dropbox, or Spotify, the client often plays a critical role in the overall design, enabling features like offline functionality, reducing server load, and improving user experience.

So how does this work in practice?

1. Just like in our high-level design, we use on-device location services to record the user's GPS coordinates at a constant interval.
2. We record this event data locally on the device in an in-memory buffer (e.g., an array of GPS coordinate and timestamp pairs).
3. To prevent data loss in case of device shutdown or battery depletion, we'll periodically persist this buffer to the device's local storage every ~10 seconds:
 - For iOS: We can use Core Data for larger, structured datasets, or UserDefaults for smaller amounts of simple key-value data.
 - For Android: We can use Room database for larger, structured datasets, or SharedPreferences for smaller amounts of simple key-value data.
4. When the app is reopened or the activity is resumed, we first check local storage for any saved data and load it into our in-memory buffer before continuing to record new data.
5. Once the activity is complete and the device is online, we send all the accumulated data to our server in a single request. For very long activities, we might need to implement a chunking strategy to handle large amounts of data efficiently.
6. We can also implement a background sync mechanism that attempts to upload data periodically when a connection becomes available, even if the activity isn't complete yet. This balances efficiency with data durability.
7. Upon confirmation that the data was saved to our remote database, we can delete the local buffer and are ready to go again.

This approach ensures that even if the device unexpectedly shuts down, we'll only lose a maximum of 10 seconds of activity data (or whatever interval we choose for our periodic saves). However, it's important to balance the frequency of GPS tracking and data persistence with battery life considerations.



How can we support tracking activities while offline?



The exact implementation details may vary depending on the mobile platform and the size of the data being stored. For larger datasets, using a local SQLite database might be more appropriate than options like UserDefaults or SharedPreferences. But the details of this are reserved more for Mobile System Design interviews, not a general SWE interview.



We're done with our first deep dive, and we still have a simple client-server architecture. Why? Especially when most of our other breakdowns use microservices? The truth is, there is no one right answer here. But in this case, we offloaded so much of the write load to the client that we don't have a ton for our backend to do. What we do have still needs to scale, and we will get to that in a bit, but we don't need separate microservices to scale independently given no significant read/write skew across different paths. It could be argued that this will make it harder for the engineering team to grow, with everyone touching the same codebase, but if Meta can famously manage it, so can we.

2) How can we scale to support 10 million concurrent activities?

Now that we track activities locally, scaling our backend became pretty easy! We cut down on the number of requests our backend services receive by a factor of 100x since we only send data once a run completes now. Our production engineering team is going to be thrilled, so too the finance team.

We do still have some potential bottlenecks that are worth addressing.

Let's start by looking at our database. With ~100M DAU doing an activity each day we add up to ~100M new activities each day. Over a year, that's ~36500M activities or ~36.5B activities.

To estimate the amount of storage we need for each activity, we can break it down as follows:

- Basic metadata like status, userID, start/end time, etc should be pretty cheap. Let's say ~100 bytes per activity.
- The route is the most expensive part. If the average activity is 30 minutes and we take GPS coordinates, on average, every ~3 seconds, then we'll have about 600 points per activity. Each route row needs a latitude and longitude field as well as a timestamp. This is an additional $(8\text{bytes} + 8\text{bytes} + 8\text{bytes}) * 600 = \sim 15\text{KB}$.

$$15\text{KB} * 36.5\text{B} = 547.5\text{TB of data each year}$$

This is a good amount of data, but it's far from unmanageable. Here are some things we could do to handle this:

1. We can shard our database to split the data across multiple database instances. We can shard by the time that the activity was completed since the majority of our queries will want to see recent activities.

2. We could introduce data tiering. The chances that someone wants to view a run from several years ago are pretty low. To reduce storage costs, we could move older data to cheaper storage tiers:
 - Hot data (recent activities) stays in fast, expensive storage
 - Warm data (3-12 months old) moves to slower, cheaper storage
 - Cold data (>1 year old) moves to archival storage like S3
3. We can introduce caching if needed. If we find that we are frequently querying the same activities, we can introduce caching to reduce the read load on our database. This would not be a priority for me, as read throughput should be pretty low. but its the first thing we would do if load times become unacceptable.

Pattern: Scaling Writes

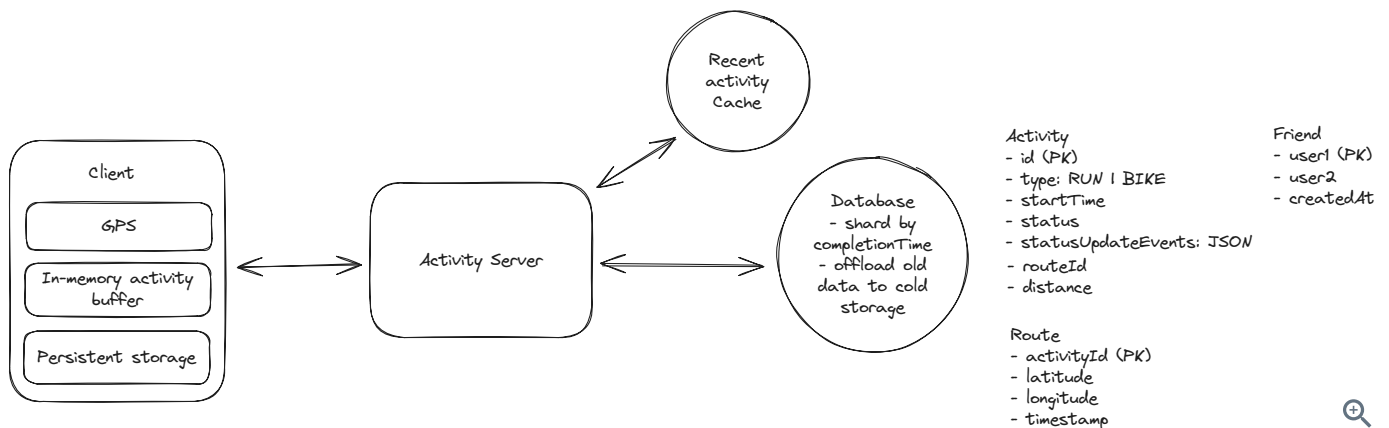
The strategies we've discussed here are all common among systems where we need to scale writes. This challenge is common to many system design challenges and it's useful to understand the pattern for instances where interviewers may want to dive deeper.

[Learn This Pattern](#)

What database should we choose?

Hot take. It doesn't really matter. This is a large but totally manageable amount of data; there is no crazy read or write throughput, and data is relational but not heavily so. Realistically, all popular major database technologies would work great here.

For the Activity Service, given both read and write throughput should be pretty low, I see no reason to break this into microservices that scale independently. Instead, when we run into issues with memory, cpu, or network limitation we can just scale the service horizontally.



How can we scale to support 10 million concurrent activities?

3) How can we support realtime sharing of activities with friends?

One common follow up question in this interview is, "what happens when you want to allow friends to follow along with activities in real-times". In this case, friends don't just see activity statistics once completed, but they can watch you mid run/bike ride -- seeing your stats and routes update in near real-time.

This moves the design closer to other popular realtime systems like [FB Live Comments](#) or [Whatsapp](#).

To enable real-time sharing with friends, we'll reintroduce periodic server updates during activities. While maintaining local device logic for user-facing data, we'll send location updates to the server every 2-5 seconds. As

the server gets these updates, they'll be persisted in the database and broadcast to all of the user's friends.

Now, I know what you're likely thinking, "Websockets!" This isn't wrong per se, you could absolutely implement this with websockets or SSE, but I'd strongly argue it's over-engineering the problem.

While you could introduce a real-time tracking service which connects to friends clients via Websocket or SSE and use pub-sub to broadcast updates to friends, this introduces a lot of unnecessary complexity.

Instead, there are two key insights that suggest a simpler, polling mechanism will be more effective:

1. Updates are predictable: Unlike with Messenger or Live Comments, we know that the next update should come in the next 2-5 seconds. This predictability allows for efficient polling intervals.
2. Real-time precision isn't critical: Friends don't need to see up-to-the-second information. A slight delay of a few seconds is acceptable and won't significantly impact the user experience.

⚡ Pattern: Real-time Updates

One of the most important things to understand about the realtime updates pattern is when it's not necessary! In our pattern breakdown we give you a simple flow chart that you can follow to determine which protocols to use and how to architecture your backend.

[Learn This Pattern](#)

Informed by these insights, we can implement a simple polling mechanism where friends' clients periodically request updates at the same interval that the athlete's device is sending updates to the server (offset by a few seconds to account for latency).

We can further enhance the user experience by implementing a smart buffering system. This approach involves intentionally lagging the displayed location data by one or two update intervals (e.g., 5-10 seconds). By buffering the data, we can create a smoother, more continuous animation of the athlete's movement. This eliminates the jarring effect of sudden position changes that can occur with real-time updates. To friends viewing the activity, the athlete will appear to be in constant motion, creating a more engaging, "live-stream-like" experience. While this approach sacrifices absolute real-time accuracy, it provides a significantly improved visual experience that better matches users' expectations of how a live tracking feature should look and feel. The intentional lag also helps compensate for any network latency, ensuring a more consistent experience across different network conditions.

4) How can we expose a leaderboard of top athletes?

Another natural extension to this problem could be to expose a leader board of the top athletes by activity type and distance. We could filter by country, region, city, etc.

Here are some approaches we could take:

Bad Solution: Naive Approach



Good Solution: Periodic Aggregation



Great Solution: Real-time Leaderboard with Redis



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Sort By
Popular **A** Aingkaran Jega

★ Top 5% • 1 year ago • edited 11 days ago

for the last deep dive you have the leader board for the countries in the redis sorted set as :
`redis.zadd(f"leaderboard:run:USA", {user_id: timestamp})`

but shouldn't it be user_id: score, I assume for this we're just making separate sets for each country so timestamp would be irrelevant here right?

21 [Reply](#)

A AppropriateGreenHoverfly664

Premium • 1 year ago

If user1 is primaryKey in the friends table how will the user have more friends?
Will it not be always just 1. I'm bit confused here

9 [Reply](#)

E Eric Tsend

Premium • 10 months ago

I thought it was duplicated for each relationship which seems redundant and costly, but shouldn't that take care of your concern here?

1 [Reply](#)

S socialguy

★ Top 1% • 11 months ago • edited 2 days ago

1. The Friend table can't have `user1` as the PK, since `user1` may have multiple friends. It needs to be a composite key `(user1, user2)`.
2. In HLD 2.2, the coordinates should be sent to `/activities/:activityId/routes` as previously outlines during API Design.

3. In Deep Dive 3, filtering by time, what happens if the user does more than one activity in the last 24 hours? If the user id is the member, the sorted set wouldn't be able to handle that.

👍 9 [Reply](#)

H HistoricalAzureCuckoo704

Premium • 11 months ago

re 1. I think this also goes for the route table. Can't have the activity_id as the PK, needs a route_id as PK and possibly set the activity_id as the Foreign key

👍 1 [Reply](#)

K kstarikov

★ Top 5% • 6 months ago

It needs to be a composite key (user1, user2)

This will make it difficult to run queries like 'get all friends of user1'.

👍 0 [Reply](#)

S socialguy

★ Top 1% • 11 months ago

The last point is moot, the sorted set member is the activity id, which is unique. I got confused since the same section shows sorted set by user id.

👍 0 [Reply](#)

P pssharma1699

Premium • 11 months ago • edited 4 days ago

Thanks for the explanation, I am little confused about the database choice here,

lets say there are 100M DAU and 600 route related updates are being sent by per user per day that transforms to 60k updates/sec (also this is considering there is no peak load and every 3 seconds we will send route update to server).

Handling this doesn't seem to be very straight forward. We would need some kind of write optimised database.

This problem might be solved by using the batched approach where we are batching updates in client and sending at once to server (even if we sent it per 10 mins the write load is 1000 updates/sec) which is edge limit of postgres.

Batching approach will not work in case we are planning to send these updates in realtime to friends as then we would need updates to be sent at regular interval.

One solution I can think of is:

1. Create a different route service (Rationale being that the traffic pattern is different between Activity and Route service) and use cassandra to store the routes which can handle the heavy load.

👍 6 [Reply](#)

F FancyTanGoldfish347

Premium • 11 months ago

I think this is quite a good point to dive into, especially when it come to real machine count + write ops benchmarks from different sources. Of courses there are many factors that play into this (all DBs can eventually scale to this traffic demand if cost and hardware was infinite), but DBs like Cassandra would be able to scale its # of write ops with fewer machines making it more efficient. <https://www.linkedin.com/pulse/nosql-mongodb-vs-cassandra-shrutika-poyrekar>

But, we don't really need to worry about such high QPS since it sounds like the design mentioned doing direct writes to Redis for live activities since it is considered the "hot data". We would need to consider a cadence for flushing this data to the slower DB we decide to choose. Using Cassandra doesn't seem necessary based on this design decision alone.

👍 1 [Reply](#)

**udit agrawal**

★ Top 10% • 5 months ago • edited 5 months ago

Can cassandra serve for large read throughput or what can be a good approach?

I feel we can even use Redis here to put the coordinate data as redis can also support such high throughput, and we can push the coordinate data to other persistent DB(such as postgres) in batches for the case of later view of route.

 0 [Reply](#)**A****Aditya Jain****Premium** • 9 months ago • edited 7 months ago

Wouldnt it be wise to shard the database by activity_id (instead of time) , because it may become hot shard if many users are online at the same time? I think we should shard by activity id and then partition it by time.

 6 [Reply](#)**F****FinancialGreenTick391**

• 8 months ago • edited 8 months ago

On one hand, It feels like "the majority of our queries will want to see recent activities" is a good reason not to shard by timestamp, since many queries will hit a specific shard.

On the other hand, when fetching the (chronological) feed, we don't want to hit multiple shards.

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Unit #4238 Seattle

WA 98103

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